

EEE-1221

Electronics

Course teacher:
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Crystal diode Rectifier

Rectification means converting AC signal into DC signal

Crystal diode converts AC signal into pulsating DC signal

Pulsating DC– Filtering– Constant DC

Tow types of rectifier circuits

- a) Half wave rectifier
- b) Full wave rectifier

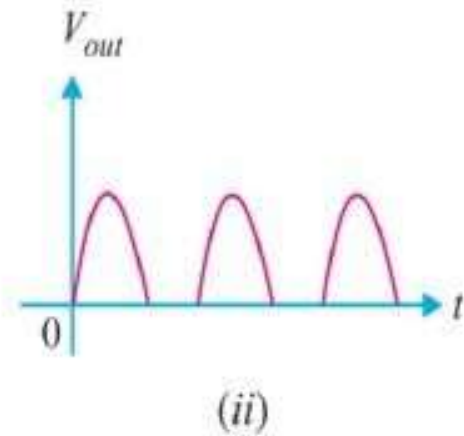
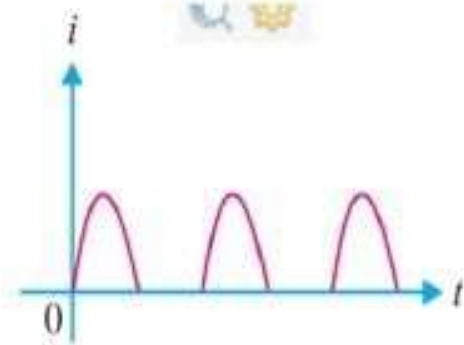
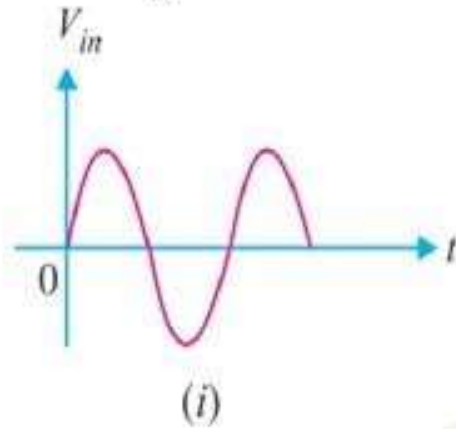
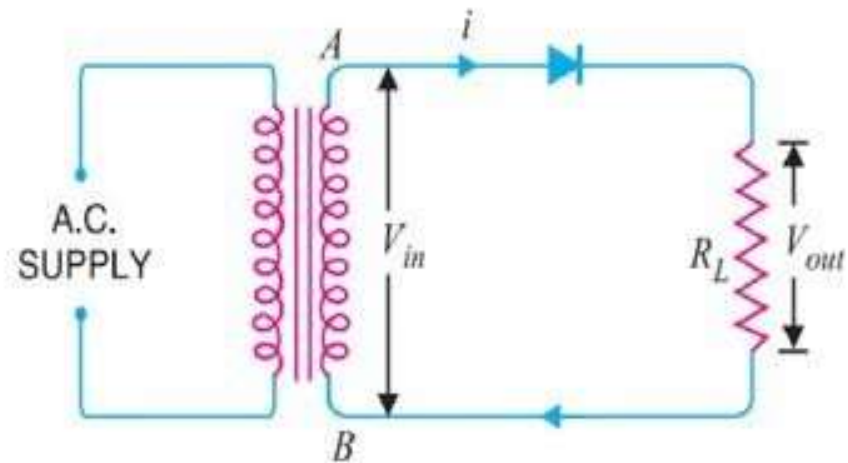
Half wave Rectifier(Clipper circuit)

In half-wave rectification, the rectifier conducts current only during the positive half-cycles of input ac supply.

The negative half-cycles of ac supply are suppressed i.e. during negative half-cycles, no current is conducted and hence no voltage appears across the load. Therefore, current always flows in one direction (i.e. dc) through the load though after every half-cycle.

“That’s why diode is a unidirectional device”

Half wave Rectifier(Clipper circuit)



Disadvantages of Half wave Rectifier

The main disadvantages of a half-wave rectifier are :

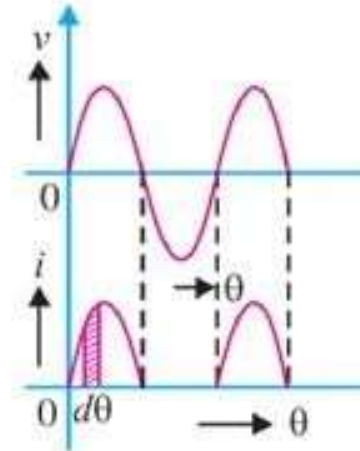
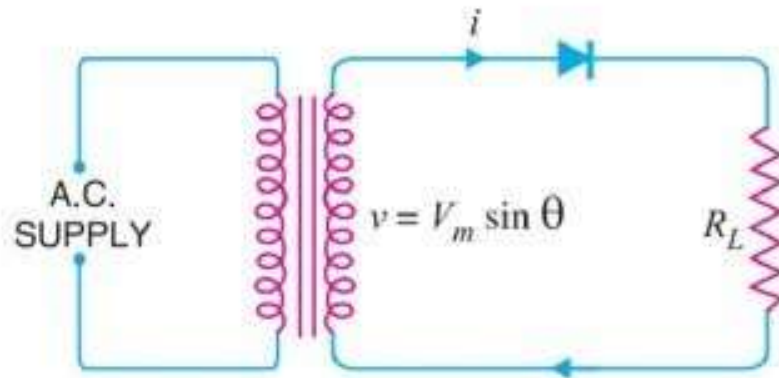
- (i) The pulsating current in the load contains alternating component whose basic frequency is equal to the supply frequency ($f_{out} = f_{in}$). Therefore, an elaborate filtering is required to produce steady direct current.
- (ii) The a.c. supply delivers power only half the time. Therefore, the output is low.

Efficiency of Half wave Rectifier

The ratio of d.c. power output to the applied input a.c. power is known as rectifier efficiency.

$$\eta = \frac{\text{DC output power}}{\text{input ac power}}$$

Diode conducts and supplies power to the load only during positive half cycle of the input ac supply.



Output dc power is pulsating dc. So for output dc power we need to find the average load current.

Efficiency of Half wave Rectifier

Let voltage at the secondary side of the transformer, $v = v_m \sin \theta$

Resistances of the diode and the load are r_f and R_L

So the average current,

$$\begin{aligned} I_{av} &= \frac{1}{2\pi} \int_0^\pi \frac{v}{R} d\theta \\ &= \frac{1}{2\pi} \int_0^\pi \frac{v_m \sin \theta}{R_L + r_f} d\theta \\ &= \frac{v_m}{2\pi(R_L + r_f)} \int_0^\pi \sin \theta d\theta \\ &= \frac{v_m}{2\pi(R_L + r_f)} * 2 \\ &= \frac{v_m}{\pi(R_L + r_f)} = \frac{I_m}{\pi} \end{aligned}$$

$$(here, I_m = \frac{v_m}{(R_L + r_f)})$$

Efficiency of Half wave Rectifier

So output dc power, $P_{dc} = I_{av}^2 * R_L = \left(\frac{I_m}{\pi}\right)^2 * R_L$

Input ac power, $P_{ac} = I_{rms}^2 * (R_L + r_f) = \left(\frac{I_m}{2}\right)^2 * (R_L + r_f)$

$$\left(\text{here, } I_{rms}(\text{for half cycle}) = \sqrt{\frac{1}{2} \times \frac{1}{2\pi} \int_0^{2\pi} \left\{ \frac{v_m \sin \theta}{(R_L + r_f)} \right\}^2 d\theta} \right) = \frac{I_m}{2}$$

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{\left(\frac{I_m}{\pi}\right)^2 * R_L}{\left(\frac{I_m}{2}\right)^2 * (R_L + r_f)} = \frac{0.406 R_L}{(R_L + r_f)} = \frac{.406}{1 + \frac{r_f}{R_L}} = .406 \times 100\% = 40.6\%$$

$r_f \ll R_L, \text{ so, } 1 + \frac{r_f}{R_L} \approx 1$

Max. rectifier efficiency = 40.6%

This shows that in half-wave rectification, a maximum of 40.6% of a.c. power is converted into d.c. power.

Full wave Rectifier

The following two circuits are commonly used for full-wave rectification:

- (i) Centre-tap full-wave rectifier
- (ii) Full-wave bridge rectifier

Full wave bridge Rectifier

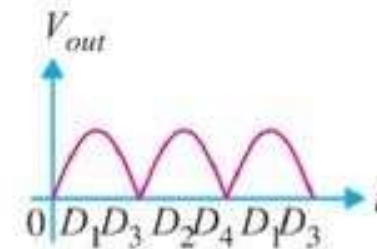
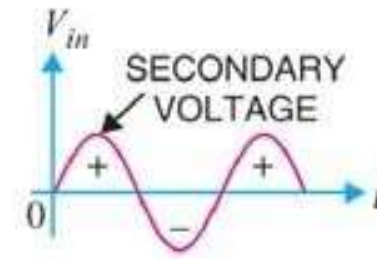
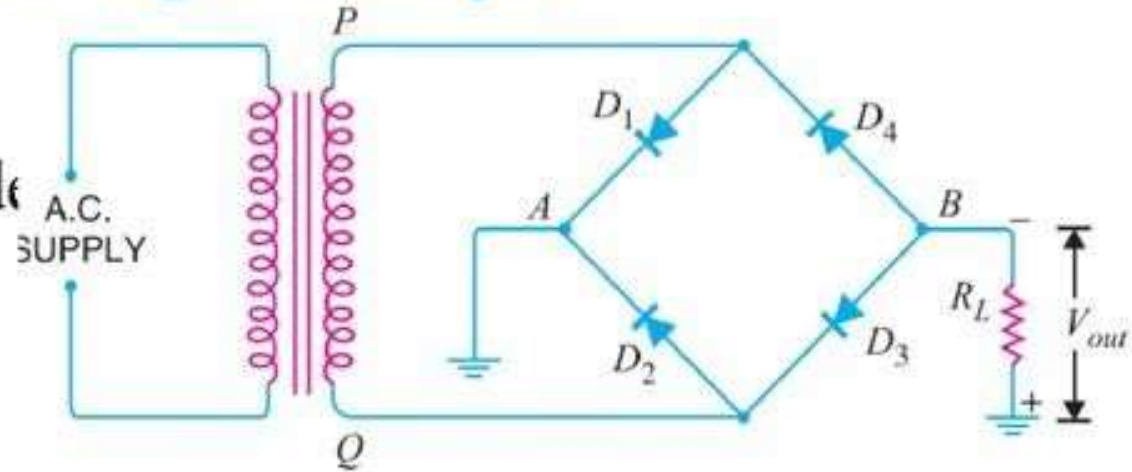
Working:

❖ During positive half cycle diodes D_1 and D_3 are forward biased.

D_2 and D_4 are reverse biased

❖ So current flows through

D_1 , R_L , D_3 and return back to the secondary winding of the transformer output voltage is obtained across the load, R_L



Full wave bridge Rectifier

Working:

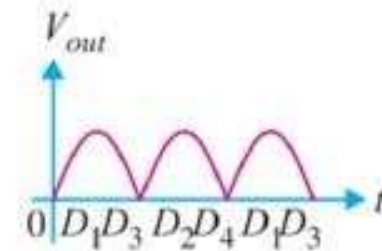
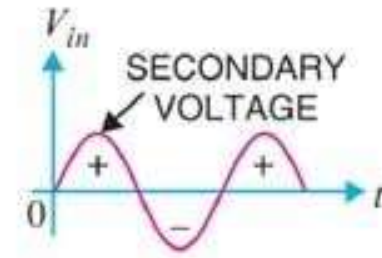
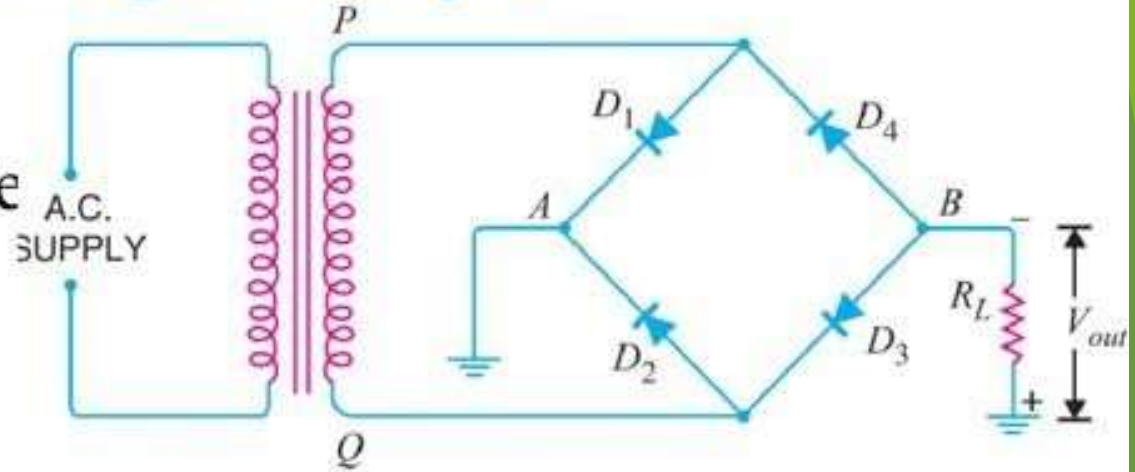
❖ During negative half cycle diode D_2 and D_4 are forward biased.

D_1 and D_3 are reverse biased

❖ So current flows through

D_2 , R_L , D_4 and return back to the secondary winding of the transformer and output voltage is obtained across the load, R_L .

So we obtain power at load for both cycle of the input ac



Full wave bridge Rectifier

Working:

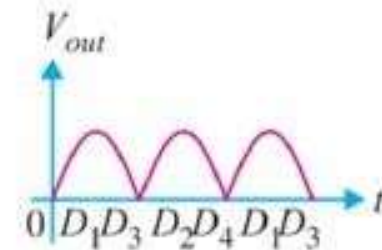
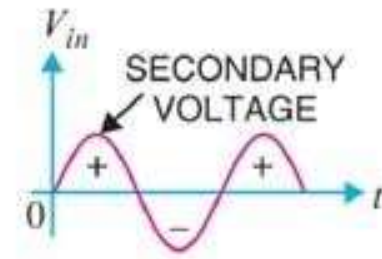
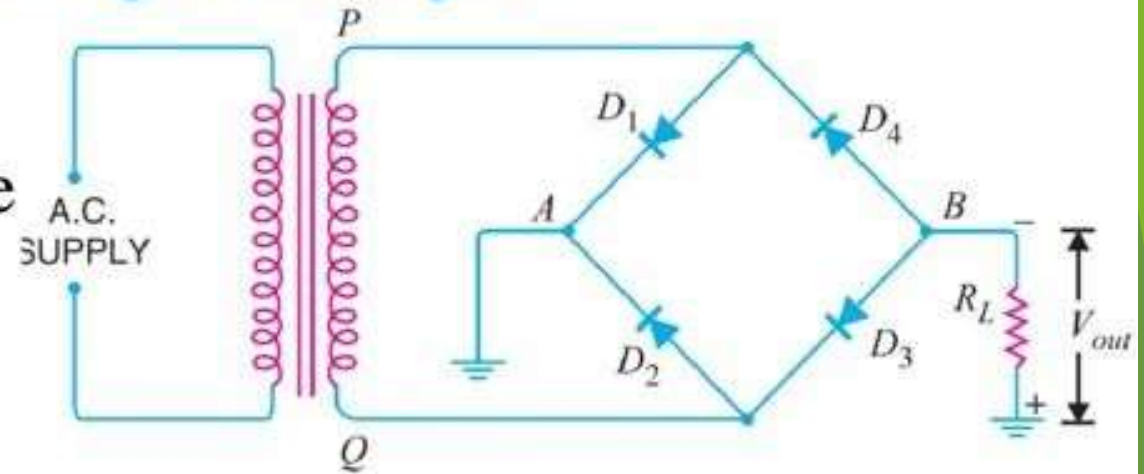
❖ During negative half cycle diode D_2 and D_4 are forward biased.

D_1 and D_3 are reverse biased

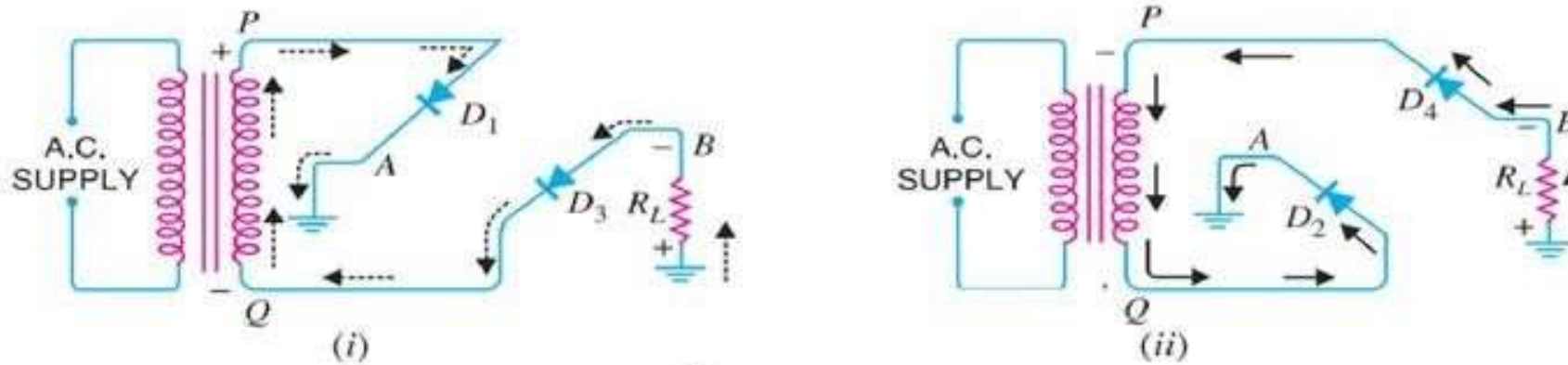
❖ So current flows through

D_2 , R_L , D_4 and return back to the secondary winding of the transformer and output voltage is obtained across the load, R_L .

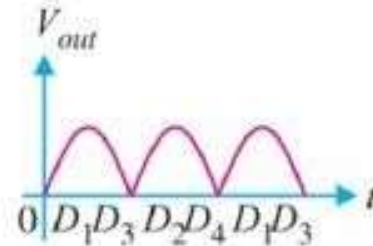
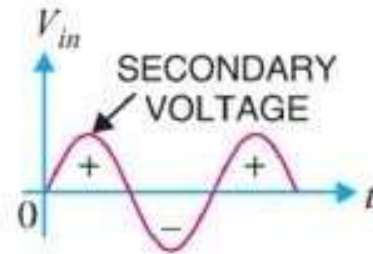
So we obtain power at load for both cycle of Input ac supply



Load Current path for positive and negative cycle of the input ac

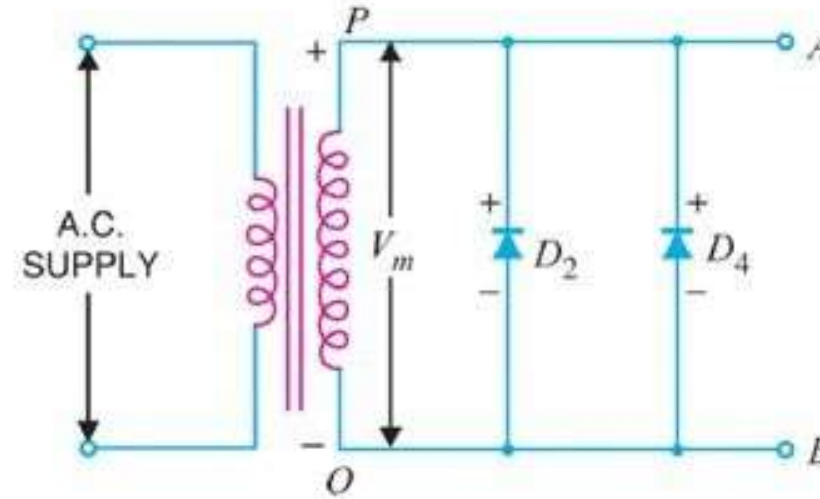
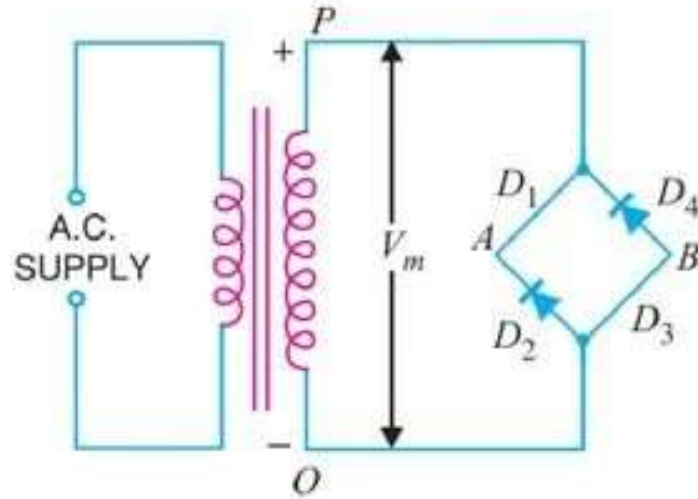


Here, $f_{out} = 2f_{in}$



Peak Inverse Voltage

The peak inverse voltage is the maximum voltage that can be applied across the diode during reverse biased condition. It must be less than the breakdown voltage of the diode.



Here, the PIV of the diodes is V_m volt.

disadvantages Full wave bridge Rectifier

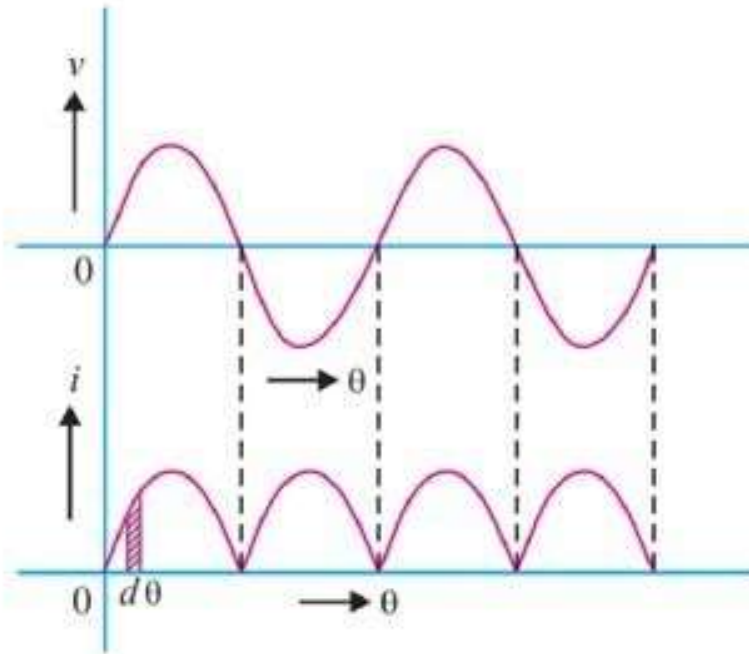
Disadvantages:

- i) It requires four diodes
- ii) As during each half-cycle of a.c. input two diodes that conduct are in series, therefore, voltage drop in the internal resistance of the rectifying unit will be twice as great as in the center tap circuit. This is objectionable when secondary voltage is small.

Efficiency of Full wave Rectifier

$$\eta = \frac{\text{DC output power}}{\text{input ac power}}$$

Diodes conduct and supplies power to the loads for full cycle of the input ac supply.



Output dc power is pulsating dc. So for output dc power we need to find the average load current.

Efficiency of Full wave Rectifier

Let voltage at the secondary side of the transformer, $v = v_m \sin \theta$

Resistances of the diode and the load are r_f and R_L

So the average current,

$$\begin{aligned} I_{av} &= \frac{1}{\pi} \int_0^\pi \frac{v}{R} d\theta \\ &= \frac{1}{\pi} \int_0^\pi \frac{v_m \sin \theta}{R_L + r_f} d\theta \\ &= \frac{v_m}{\pi(R_L + r_f)} \int_0^\pi \sin \theta d\theta \\ &= \frac{v_m}{\pi(R_L + r_f)} * 2 \\ &= \frac{2v_m}{\pi(R_L + r_f)} = \frac{2I_m}{\pi} \end{aligned}$$

$$(here, I_m = \frac{v_m}{(R_L + r_f)})$$

Efficiency of full wave Rectifier

So output dc power, $P_{dc} = I_{av}^2 * R_L = \left(\frac{2I_m}{\pi}\right)^2 * R_L$

Input ac power, $P_{ac} = I_{rms}^2 * (R_L + r_f) = \left(\frac{I_m}{\sqrt{2}}\right)^2 * (R_L + r_f)$

$$\left(\text{here, } I_{rms}(\text{for full cycle}) = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} \left\{ \frac{v_m \sin \theta}{(R_L + r_f)} \right\}^2 d\theta} \right) = \frac{I_m}{\sqrt{2}}$$

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{\left(\frac{2I_m}{\pi}\right)^2 * R_L}{\left(\frac{I_m}{\sqrt{2}}\right)^2 * (R_L + r_f)} = \frac{0.812R_L}{(R_L + r_f)} = \frac{.812}{1 + \frac{r_f}{R_L}} = .812 \times 100\% = 81.2\%$$
$$r_f \ll R_L, \text{ so, } 1 + \frac{r_f}{R_L} \approx 1$$

Max. rectifier efficiency = 81.2%

This is double the efficiency due to half-wave rectifier. Therefore, a full-wave rectifier is twice as effective as a half-wave rectifier.